AP-7: Home Perimeter Erosion

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Attribution

> This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

Abstract

> *Home Perimeter Erosion (AP-7) is the first anti-pattern in Category 5 of the Phase D3 taxonomy: Perimeter and Content Failures. It names the failure mode in which a Self's participation in a Full Aspect Integration (FAI) event produces unintended changes to the Self's home governance architecture — unauthorized modifications to home DNA, reduced home governance authority over certain content, or involuntary exposure of non-contributed aspects to other participating Selves. The architectural commitment AP-7 violates is the additive property of Paper 3 Claim 1: the inter-Self perimeter must add a new governance scope without subtracting from the home perimeter. When that property fails, FAI participation has not extended the Self's coordination reach — it has substituted for part of the Self's home governance, eroding the perimeter it was designed to leave intact. AP-7 is almost always a downstream consequence of AP-4 (automatic DNA absorption) or AP-6 (implicit contribution scope), making co-occurrence identification the first diagnostic step. The override right, available without justification under Paper 1 Claim 3, provides the immediate remediation mechanism for removing unauthorized home substrate changes; root-cause investigation then addresses the originating anti-pattern to prevent recurrence.*

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1. Anti-Pattern Name and Category

Name: AP-7 — Home Perimeter Erosion

Category: 5 — Perimeter and Content Failures (opening anti-pattern)

Series position: D3.18, note #593 in the derivation series. AP-7 is the first of four Category 5 anti-patterns and the eighteenth Phase D3 note overall.

Primary commitment violated: Paper 3 Claim 1, D1.03 — the inter-Self perimeter's additive property.

2. Description

Paper 3 Claim 1 establishes that the shared substrate spans multiple Selves' home governance perimeters as an *additive* architectural move. The inter-Self scope adds a new, temporary governance space — the shared substrate — in which two or more Selves coordinate under joint authority. The critical property of this addition is that it leaves each participating Self's home governance perimeter intact. FAI participation extends what a Self can do by giving it access to the inter-Self coordination layer; it does not reduce what the Self's home governance controls within its own perimeter. This is the additive property: the shared substrate adds a scope on top of the home perimeter, not in place of any part of it.

Home Perimeter Erosion names the failure in which the addition has become a substitution. The inter-Self FAI event has not merely opened a new coordination scope — it has reached into the home perimeter and modified it in ways that governance did not authorize. The Self that participated in the FAI event discovers afterward that its home governance architecture is different from what it was before participation. Something the Self governed before the FAI event is now outside its governance reach, or its home substrate now carries content it did not authorize, or aspects the Self did not intend to contribute have been exposed to the other participating Self. In each case, the inter-Self perimeter has subtracted from the home perimeter rather than only adding above it.

Three manifestations of erosion are architecturally distinct, though they share the same underlying violation:

> **Home DNA changes without governance authorization.** The Self's home DNA — the governance-authorized rules and structured content that constitute the Self's identity and operational memory — has been modified during or after FAI participation without going through directed selection, the governance mechanism that authorizes DNA evolution. The modifications trace to FAI participation rather than to any home-authorized governance event. The Self's home substrate now carries rules or content that governance did not choose.
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> **Reduced home governance authority over certain operations.** Home governance practitioners discover they can no longer exercise one or more of the three rights — inspect, modify, override — over certain home substrate content they previously governed. The content exists in the home substrate, but the governance authority over it has been encumbered or made ambiguous by the FAI event. In the most severe form, the content has become effectively jointly governed by both Selves, even though the participating Self's governance never authorized joint authority over that content.
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> **Involuntary exposure of non-contributed aspects.** The Self contributed specific aspects to the shared substrate, but other aspects — not part of the intended contribution — have also appeared in the shared substrate and become visible to the other participating Self. Home content that governance did not authorize for sharing has crossed the inter-Self boundary. Even if the home perimeter has not been structurally modified, the effective scope of the home perimeter has shrunk: content that was inside the home perimeter before the FAI event is now also outside it.

All three manifestations share the underlying structure: the inter-Self scope has not remained additive. It has reached through the home boundary and produced effects inside the home perimeter that governance did not authorize.

3. Detection Criteria

Four detection signals indicate that home perimeter erosion has occurred or is occurring. Each corresponds to one of the D2.17 home perimeter integrity checks.

Home DNA differences coinciding with FAI participation, without governance authorization. A post-event comparison of home DNA — before and after FAI participation — reveals differences that cannot be attributed to directed selection events the Self's governance authorized. The timing alignment with FAI participation is the diagnostic signal. DNA changes that trace to home-authorized directed selection are expected; changes that trace to FAI participation without corresponding directed-selection records are the erosion signal.

Non-contributed aspects' content appearing in the shared substrate without contribution records. The shared substrate contains content from the Self that goes beyond the aspects governance configured for contribution. No contribution record for those aspects exists; the content appears in the shared substrate without having passed through the governed contribution pathway.

Home governance practitioners reporting inability to exercise the three rights over previously governed content. Practitioners with home governance authority discover that certain home substrate content is no longer fully under their governance control — that they cannot inspect it in its current state, cannot modify it without navigating an inter-Self approval process, or cannot override it without effects on the other participating Self's substrate.

Post-event comparison finding home DNA differences not attributable to authorized events. The D2.17 Check A comparison is the canonical detection mechanism: compare home DNA state immediately before FAI participation with the state immediately after, and check each difference against the home-authorized directed-selection record. Any difference without a corresponding directed-selection record is a home perimeter erosion indicator.

4. Governance Commitment Violated

Primary commitment: Paper 3 Claim 1, D1.03 — the inter-Self perimeter's additive property.

Paper 3 Claim 1 commits that the shared substrate spans participating Selves' home governance perimeters without replacing them. The perimeter is additive: the inter-Self scope adds above the home perimeter. Home perimeter erosion violates this commitment directly. The erosion is not a marginal deviation from the additive property — it is its negation. When the inter-Self scope modifies home substrate content without governance authorization, the FAI event has substituted for part of the Self's home governance: the shared substrate has not only added an inter-Self scope but has reached into the home scope and changed it. Addition has become substitution.

Secondary commitment: Paper 1 Claim 3 — human-governed authority.

Paper 1 Claim 3 establishes that humans retain the three rights — to inspect, modify, and override substrate content and orchestration rules — at all times during the substrate's existence. The commitment is architectural and temporal: the rights must be available as a property of the system's design, at any time, not only at predetermined checkpoints. Home perimeter erosion violates Paper 1 Claim 3 at the home perimeter scope because authority over home substrate content has been diminished during FAI participation. A Self whose home DNA has been modified without authorization, whose governance authority over certain operations has been reduced, or whose home content has been involuntarily shared cannot fully exercise the three rights over its own home substrate.

Operational reference: D2.17 — home perimeter integrity.

D2.17 formalizes four home perimeter integrity properties and the three governance checks that verify them. Home perimeter erosion violates one or more of the four integrity properties. The D2.17 checks are the detection mechanism; the four properties are the specification of what must be intact for the home perimeter to remain additive.

5. Consequences

The consequences of home perimeter erosion distribute across four areas of governance operation.

> **Home governance sovereignty is compromised.** The Self's governance architecture is partly outside its governance control after FAI participation. The Self governs its home substrate — but the home substrate now includes content it did not authorize and possibly lacks content its governance previously controlled. Governance sovereignty requires knowing what is inside the perimeter; erosion breaks that knowledge.
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> **The authorization chain is broken.** The four-link authorization chain (D2.67) that connects home substrate changes to human-authorized governance events is broken wherever erosion has occurred. Home substrate changes trace to FAI participation rather than to directed selection or other home-authorized governance mechanisms. The traceability property fails for the eroded content.
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> **Future FAI contributions may include unintended content.** A Self with an eroded home perimeter cannot accurately predict what a future FAI event will contribute to the shared substrate. If non-contributed aspects have been modified by erosion, or if the contribution scope is now implicitly broader than governance configured, the next FAI event will surface aspects the Self did not intend to share. Erosion in one FAI event degrades the governance integrity of all subsequent FAI events involving the same Self.
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> **Home governance trust is undermined.** Home governance practitioners exercise authority on the assumption that what they govern is what they think they govern. An eroded home perimeter breaks that assumption. Practitioners cannot verify — without performing the D2.17 post-event checks after every FAI event — that the home substrate they are governing is the home substrate they have been governing.
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> **Exit cannot cleanly restore the pre-FAI governance state.** D2.46 formalizes the conditions for clean exit from a FAI relationship. Clean exit requires that the home perimeter can be restored to its pre-FAI state. If erosion has blurred the home perimeter, clean exit is unavailable: the boundaries cannot be cleanly re-drawn because they were never cleanly maintained during participation.

6. Intra-Self Analog

Paper 2 establishes a three-level structure within a single Self: cells, aspects, and the Self as a whole. Each level has its own governance perimeter, and operations at one level are supposed to respect the governance boundary of adjacent levels. An operation within a cell that modifies aspect-level content without aspect-level governance authorization is a boundary violation at the cell-aspect scope. An operation within an aspect that modifies Self-level content — the DNA layer of the Self — without Self-level governance authorization is a boundary violation at the aspect-Self scope.

Home perimeter erosion is the inter-Self scope version of the same failure mode. The failure structure is identical: an operation in one scope (the inter-Self scope) produces effects in an adjacent scope (the home scope) without governance authorization from the affected scope's governance authority. The boundary that was supposed to contain the operation's effects did not contain them.

The CKS architecture maintains governed boundaries at every scope: the cell boundary within an aspect, the aspect boundary within a Self, the home perimeter at the Self's outer edge, and the inter-Self perimeter at the outermost coordination scope. Erosion at any of these boundaries is the same failure mode: an operation authorized within one scope has produced effects that required separate governance authorization in an adjacent scope, and that authorization was not obtained. Home perimeter erosion is not a new failure type introduced by inter-Self coordination — it is the application of a failure type the architecture already recognizes at smaller scopes to the larger scope Paper 3 introduces.

7. Co-Occurrence Patterns and Resolution

Co-occurrence patterns

AP-7 is almost always a downstream consequence of another anti-pattern rather than an independently arising failure. The two primary co-occurrence patterns are the diagnostic starting points for any erosion investigation.

> **AP-4 co-occurrence (automatic DNA absorption).** AP-4 names the failure in which FAI participation causes automatic changes to the Self's home DNA without directed-selection authorization. Automatic DNA absorption is the most direct mechanism of home perimeter erosion: the FAI event's DNA integration pathway reaches into the home DNA without passing through the directed-selection governance checkpoint. When AP-7's detection criteria fire — specifically, home DNA differences coinciding with FAI participation without governance authorization — AP-4 is the first co-occurrence to investigate. The mechanism: FAI participation triggered automatic DNA absorption (AP-4), which modified home DNA without directed-selection authorization, which constitutes home perimeter erosion (AP-7).

> **AP-6 co-occurrence (implicit contribution scope).** AP-6 names the failure in which the FAI configuration does not explicitly specify the contribution scope, leaving it implicit and potentially broader than governance intended. When contribution scope is implicit and over-broad, non-contributed aspects appear in the shared substrate — the involuntary exposure form of erosion. When AP-7's detection criteria fire — specifically, non-contributed aspects appearing in the shared

substrate without contribution records — AP-6 is the first co-occurrence to investigate. The mechanism: implicit FAI configuration set an unintended contribution scope (AP-6), which caused non-contributed aspects to be contributed, which exposed home content governance had not authorized for sharing (AP-7).

Less commonly, AP-7 may arise from a FAI implementation defect: a technical error in shared substrate construction that directly modifies home substrate content rather than correctly operating only within the shared substrate scope. This third causal pathway does not involve another named anti-pattern; it is diagnosed when neither AP-4 nor AP-6 is found to be present.

Resolution

Resolution proceeds in two phases: immediate restoration of sovereignty, and root-cause remediation.

Immediate restoration — the override right. The override right is structural: it requires no justification and is available at any time during the substrate's existence. If FAI participation has produced unauthorized changes to the home substrate, governance does not need to negotiate with the shared substrate, the other participating Self, or any orchestration layer. The override right applies to home substrate content as home substrate content; the unauthorized changes are, precisely, home substrate content; the right is therefore sufficient for immediate remediation. Governance exercises the override right to remove or reverse unauthorized home DNA changes, to restore authority over content whose governance had been encumbered, and to remove non-contributed aspect content that appeared without authorization. This restores governance sovereignty over the home perimeter without requiring root-cause investigation to be complete first. Sovereignty restoration and root-cause investigation are sequential, not contingent.

Root-cause remediation — investigation and repair. After sovereignty is restored, a governance audit identifies the full scope of erosion effects: which home DNA content was modified, which non-contributed aspects were exposed, and which governance rights were encumbered. The investigation then identifies the mechanism — AP-4, AP-6, or implementation defect — and addresses the root cause to prevent recurrence. The D2.17 home perimeter integrity checks, performed promptly after every future FAI event, are the ongoing prevention mechanism.

Prevention — D2.17 checks as routine. The most effective prevention is making the D2.17 home perimeter integrity verification a routine governance practice after every FAI event, not a forensic response to discovered erosion. When the checks are performed promptly after each event, the detection window is narrow, the erosion effects are limited to a single event, and the override right can restore sovereignty with minimal disruption. When the checks are deferred or skipped across multiple FAI events, erosion accumulates silently and the remediation burden multiplies.

8. Conclusion

Home Perimeter Erosion (AP-7) names the point at which FAI participation crosses from coordination to corruption of the home governance architecture. The architectural commitment Paper 3 Claim 1 makes — that the inter-Self perimeter is additive, not substitutive — is the commitment AP-7 violates. When that commitment holds, FAI participation expands a Self's coordination reach without costing anything from its home governance. When it fails, the Self has paid for inter-Self coordination with part of its home governance sovereignty, and the payment was not authorized.

The note opens Category 5 because perimeter integrity is the condition under which all other governance guarantees hold. A Self with an eroded home perimeter cannot make reliable governance commitments about anything its home substrate contains. The three rights — to inspect, modify, and override — remain available as structural commitments; but they cannot be exercised reliably over a home substrate whose boundaries have been silently modified. Restoring the home perimeter is therefore the precondition for restoring the force of every other governance commitment the Self holds.

End of D3.18 — AP-7: Home Perimeter Erosion.